

Traffic Density Estimation Using CSRNet

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Abstract—Traffic congestion remains a significant challenge in urban planning and management. This project implements a deep learning approach for accurate traffic density estimation using CSRNet (Congested Scene Recognition Network). By leveraging dilated convolutional layers, the model captures multi-scale contextual information while maintaining high-resolution feature maps. Unlike traditional regression-based counting, our approach generates a spatial density map that preserves the geometric structure of the traffic scene. The system is trained on the TRANCOS dataset, utilizing ground-truth density maps generated via Gaussian kernels. Experimental results demonstrate that the dilated architecture effectively handles heavy occlusion and perspective distortion, providing robust vehicle localization under varying environmental conditions.

Index Terms—Deep Learning, Traffic Density Estimation, CSRNet, Dilated Convolution, Computer Vision, TRANCOS Dataset, VGG-16.

I. INTRODUCTION

Real-time traffic monitoring is crucial for intelligent transportation systems (ITS). Traditional vehicle counting methods often rely on detection frameworks like YOLO or Faster R-CNN. However, these methods degrade significantly in highly congested scenarios where vehicle overlaps and perspective distortions are prevalent [1]. To mitigate these issues, density estimation methods have emerged as a superior alternative, treating the count as an integral over a spatial distribution [5].

This project addresses these limitations by employing CSRNet. By transforming the counting problem into a pixel-wise density estimation task, the model provides a granular understanding of traffic distribution. This is particularly effective for the TRANCOS dataset, which features challenging highway scenarios with low-angle perspectives and high vehicle density.

II. METHODOLOGY

A. Architecture Overview

The CSRNet model is a dual-component architecture. The **front-end** utilizes the first 10 layers of VGG-16, pre-trained on ImageNet, to extract high-level semantic features [3]. The **back-end** is a dilated convolutional neural network that replaces standard pooling layers to expand the receptive field while keeping the output feature map size consistent (Fig. 1).

B. Dilated Convolution and Receptive Field

Standard convolutions reduce spatial resolution through pooling, which loses the precise location of small vehicles in

Configurations of CSRNet			
A	B	C	D
input(unfixed-resolution color image)			
front-end (fine-tuned from VGG-16)			
conv3-64-1			
conv3-64-1			
max-pooling			
conv3-128-1			
conv3-128-1			
max-pooling			
conv3-256-1			
conv3-256-1			
conv3-256-1			
max-pooling			
conv3-512-1			
conv3-512-1			
conv3-512-1			
back-end (four different configurations)			
conv3-512-1	conv3-512-2	conv3-512-2	conv3-512-4
conv3-512-1	conv3-512-2	conv3-512-2	conv3-512-4
conv3-512-1	conv3-512-2	conv3-512-2	conv3-512-4
conv3-256-1	conv3-256-2	conv3-256-4	conv3-256-4
conv3-128-1	conv3-128-2	conv3-128-4	conv3-128-4
conv3-64-1	conv3-64-2	conv3-64-4	conv3-64-4
conv1-1-1			

Fig. 1. The CSRNet pipeline utilized in this work: A VGG-16 front-end for feature extraction, followed by a dilated back-end with increasing dilation rates for multi-scale density map generation.

the distance. Dilated convolution allows for a larger receptive field without downsampling [6]. A 2D dilated convolution is defined as:

$$y(i, j) = \sum_{m, n} x(i + r \cdot m, j + r \cdot n) w(m, n) \quad (1)$$

where r represents the dilation rate. As shown in Fig. 2, increasing r allows the filter to aggregate context from a wider area. Fig. 3 visualizes how this allows the network to cover standard bounding boxes of varying scales.

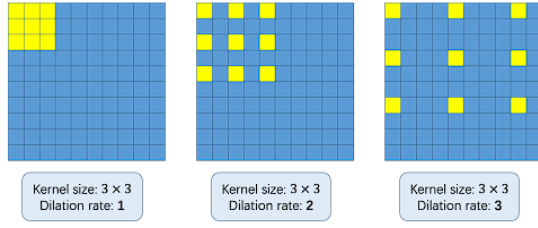


Fig. 2. Visualization of standard kernel ($r = 1$) versus dilated kernels ($r = 2, r = 3$). Stacking dilated layers expands the effective receptive field without parameter increase.

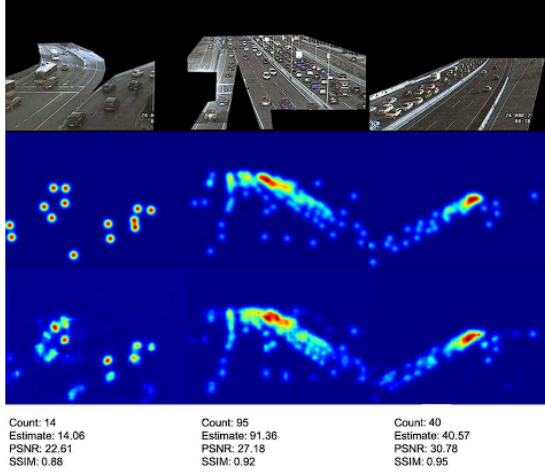


Fig. 3. A comparison of standard receptive fields vs dilated receptive fields overlaid on a traffic scene. The dilated layers allow the network to contextually define vehicle features across varying scales and extreme distortion.

C. Data Preprocessing and Augmentation

Ground-truth density maps are generated by applying a Gaussian kernel to the annotated point labels P . For each vehicle at pixel x , the density is spread using:

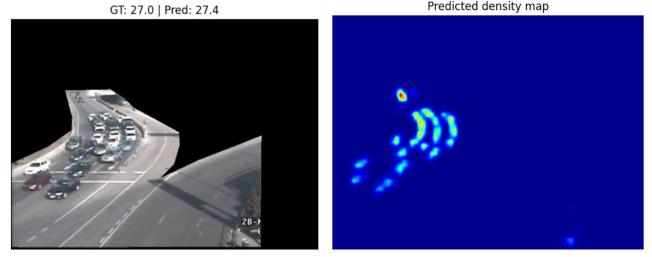
$$D(x) = \sum_{P \in \bar{P}} \delta(x - P) * G_{\sigma}(x) \quad (2)$$

To improve model generalization, we apply random cropping and horizontal flipping. Images are normalized using standard ImageNet statistics to ensure training stability [4]. Fig. 4 visualizes the results of this preprocessing step on a challenging TRANCOS sample.

III. IMPLEMENTATION DETAILS

The implementation is carried out in PyTorch.

- **Loss Function:** Mean Squared Error (MSE) is used to minimize the pixel-wise difference: $L = \frac{1}{2N} \sum ||Z(X_i; \theta) - D_i||^2$.
- **Optimization:** The Adam optimizer is used with a learning rate of $1e-5$ [4].
- **Evaluation Metrics:** Performance is measured using Mean Absolute Error (MAE) and Mean Squared Error (MSE).



(a) Input Image

(b) Ground Truth Density

Fig. 4. The data preprocessing pipeline on a high-density TRANCOS scene, showing the transformation of point annotations into a smooth, spatially preserving density map.

IV. RESULTS AND DISCUSSION

The CSRNet model demonstrates superior performance in handling the extreme congestion found in the TRANCOS dataset. The dilated layers effectively separate overlapping features, as visualized in the qualitative results grid in Fig. 5.



Fig. 5. Qualitative results comparison on different levels of congestion: (a) Input Image, (b) Ground Truth Density Map, and (c) CSRNet Predicted Density Map.

V. CONCLUSION

This project confirms that CSRNet is a highly effective architecture for traffic density estimation. Future work will investigate the use of temporal consistency in video streams to improve count stability.

REFERENCES

- [1] Y. Li, X. Zhang, and D. Chen, "CSRNet: Dilated Convolutional Neural Networks for Understanding the Highly Congested Scenes," in *Proc. CVPR*, 2018.
- [2] R. Guerrero-Sevilla et al., "TRANCOS: A dataset for vehicle counting in traffic scenes," 2015.
- [3] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," *arXiv preprint arXiv:1409.1556*, 2014.

- [4] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [5] V. Lempitsky and A. Zisserman, "Learning to Count Objects in Images," in *Advances in Neural Information Processing Systems (NIPS)*, 2010.
- [6] F. Yu and V. Koltun, "Multi-Scale Context Aggregation by Dilated Convolutions," in *ICLR*, 2016.
- [7] B. Xu et al., "Learning to Detect and Count Vehicles in Congested Scenes," *IEEE Transactions on Intelligent Transportation Systems*, 2021.